Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
AMTCA403 – RESOURCE MANAGEMENT TECHNIQUES

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. What is L.P.P?
2. Define transportation problem.
3. Define the sequencing problem.
4. What do you mean by a zero sum game?
5. Define the critical path.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Using Simplex procedure solve the L.P.P
   Maximize, $Z = 3x_1 + 2x_2$
   Subjected to, $x_1 + x_2 \le 4$
   $x_1 - x_2 \le 2 \; x_1, x_2 \ge 0$

7. Solve the following Assignment problem

		Machines				
		1	2	3	4	5
Job	1	10	11	4	2	8
	2	7	11	10	14	12
	3	5	6	9	12	14
	4	13	15	11	10	7

8. Find the optimal sequence and minimum time required to complete all the jobs for following 5 Jobs on 2 machines:

Job	1	2	3	4	5
M_1	5	1	9	3	10
M_2	2	6	7	8	4

9. Solve the following 2 × 2 game

$$\begin{matrix} & B \\ A & \end{matrix}\begin{pmatrix} 2 & 5 \\ 7 & 3 \end{pmatrix}.$$

10. Find the critical path & project duration of given below:

Activity	(1,2)	(1,3)	(1,4)	(2,5)	(3,6)	(3,7)	(4,6)	(5,8)	(6,9)	(7,8)	(8,9)
Time in month	2	2	1	4	8	5	3	1	5	4	3

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Solve the following L.P.P by Simplex method

Maximize, $Z = x_1 + 2x_2$

Subjected to, $x_1 - x_2 \geq 3$

$2x_1 + x_2 \leq 10 \;\; x_1, x_2 \geq 0$

(or)

b) Solve the following L.P.P by two phase method

Min $Z = x_1 + x_2$

Subjected to, $x_1 + x_2 \geq 0$

$2x_1 + x_2 \geq 7 \;\; x_1, x_2 \geq 0$

12. a) Find the optimum solution to the following transportation problem by MODI method

(or)

Demand		1	2	3	4	Supply
	I	21	16	25	13	11
	II	17	18	14	23	13
	III	32	27	18	41	19
		6	10	12	15	

b) Solve the Following travelling salesman problem

FROM \ TO	A	B	C	D
A	_	46	16	40
B	41	_	50	40
C	82	32	_	60
D	40	40	36	_

13. a) Find the sequence that minimizes the total elapsed time required to complete the following tasks on the machines in the order 1-2-3. Find also the minimum total elapsed time (hours) and the idle times on the machines

Task	A	B	C	D	E	F	G
Machine 1	3	8	7	4	9	8	7
Machine 2	4	3	2	5	1	4	3
Machine 3	6	7	5	11	5	6	12

(or)

b) Compute the minimum total elapsed time needed to process jobs 1and 2 on five machines $M_1, M_2, M_3, M_4 \,\&\, M_5$ given the following data.

Job 1	Sequence	M_1	M_2	M_3	M_4	M_5
	time(hrs)	2	3	4	6	2
Job 2	Sequence	M_3	M_1	M_4	M_5	M_2
	time(hrs)	4	5	3	2	6

14. a) Solve the following game using graphical method

	B_1	B_2
A_1	-6	7
A_2	4	-5
A_3	-1	-2
A_4	-2	5
A_5	7	-6

(or)

b)Use dominance property to reduce the game to2x2 game and hence find the optimal strategies and the value of the game: $\begin{bmatrix} 3 & -2 & 4 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{bmatrix}$.

15. a) The activities of a project have the following PERT time estimates.

Job	Optimistic time	Most likely time	Pessimistic time
1 - 2	3	6	15
7 – 8	4	19	28
2 - 3	6	12	30
3 - 5	5	11	17
5 - 8	1	4	7
6 - 7	3	9	27
4 - 5	3	6	15
2 - 4	2	5	8
1 - 6	2	5	14

i) Draw the network diagram & determine the critical path
ii) Find the project Completion time & its variance

(or)

b) i) What are the three main phases of a project ?
ii) Explain the dummy Activity.
iii) Write down the Fulkerson rule.

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